Final System Test Report

UART command checks for AA, BB, CC, and DD | Questa Sim-64 10.7c

Purpose

This report explains how to judge the four-command testbench and summarizes the supplied Questa transcript and waveform. A complete testbench pass is confirmed by the final PASS banner in the Transcript, with no FAIL line.

Command results

Command	UART bytes sent	Expected response	Evidence
AA - register write	AA 04 5A	No direct reply; the following BB readback should return 5A.	**PASS shown in transcript: AA write verified by BB readback.**
BB - register read	BB 04	5A	**PASS shown in transcript.**
CC - ALU with operands	CC 12 03 00	12 + 03 = 0015 (sent low byte first: 15, 00).	**PASS shown in transcript; waveform shows ALU output 0015.**
DD - ALU saved operands	DD 02	12 x 03 = 0036 (sent low byte first: 36, 00).	**Waveform shows function 2 and ALU output 0036**

Waveform evidence

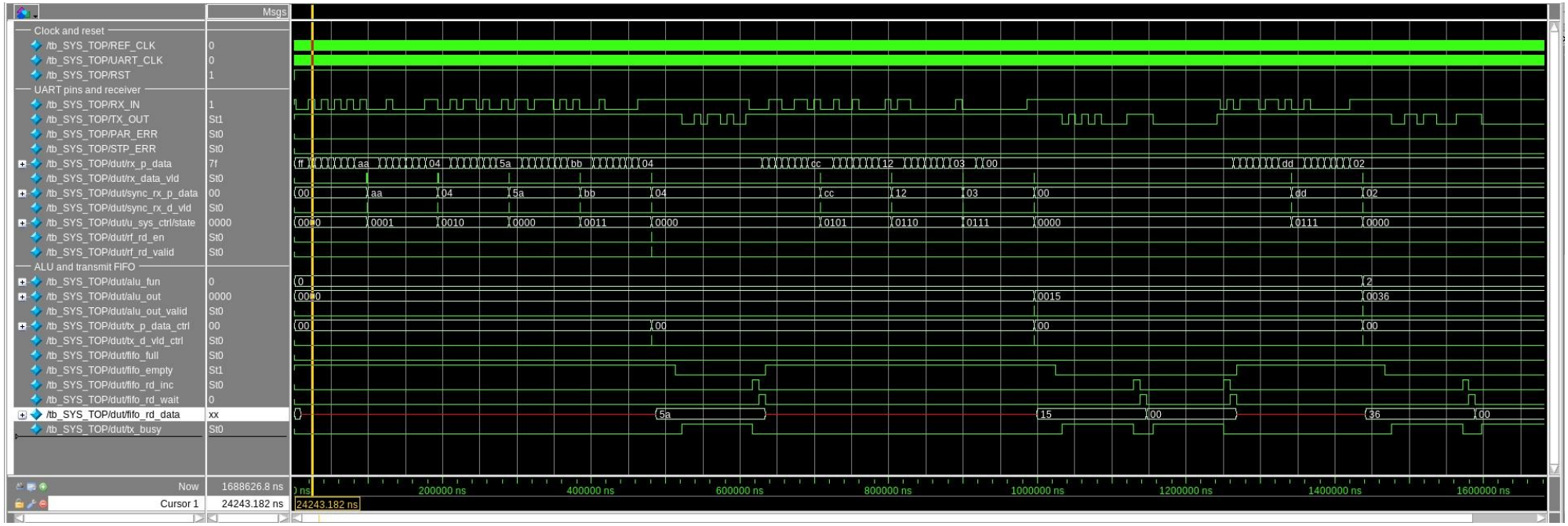


Figure 1. RX command bytes are visible in the synchronized data bus. The ALU output bus shows 0015 during CC and 0036 during DD. The FIFO read data shows response bytes low-byte first: 15, 00, then 36, 00.

Reading the trace

CC: RX receives command CC, operands 12 and 03, then function 00. The ALU output becomes 0015, the correct 16-bit sum. **DD:** RX receives DD followed by function 02. The ALU output becomes 0036, the correct product for saved operands 12 and 03. The FIFO trace shows both response bytes for each result.